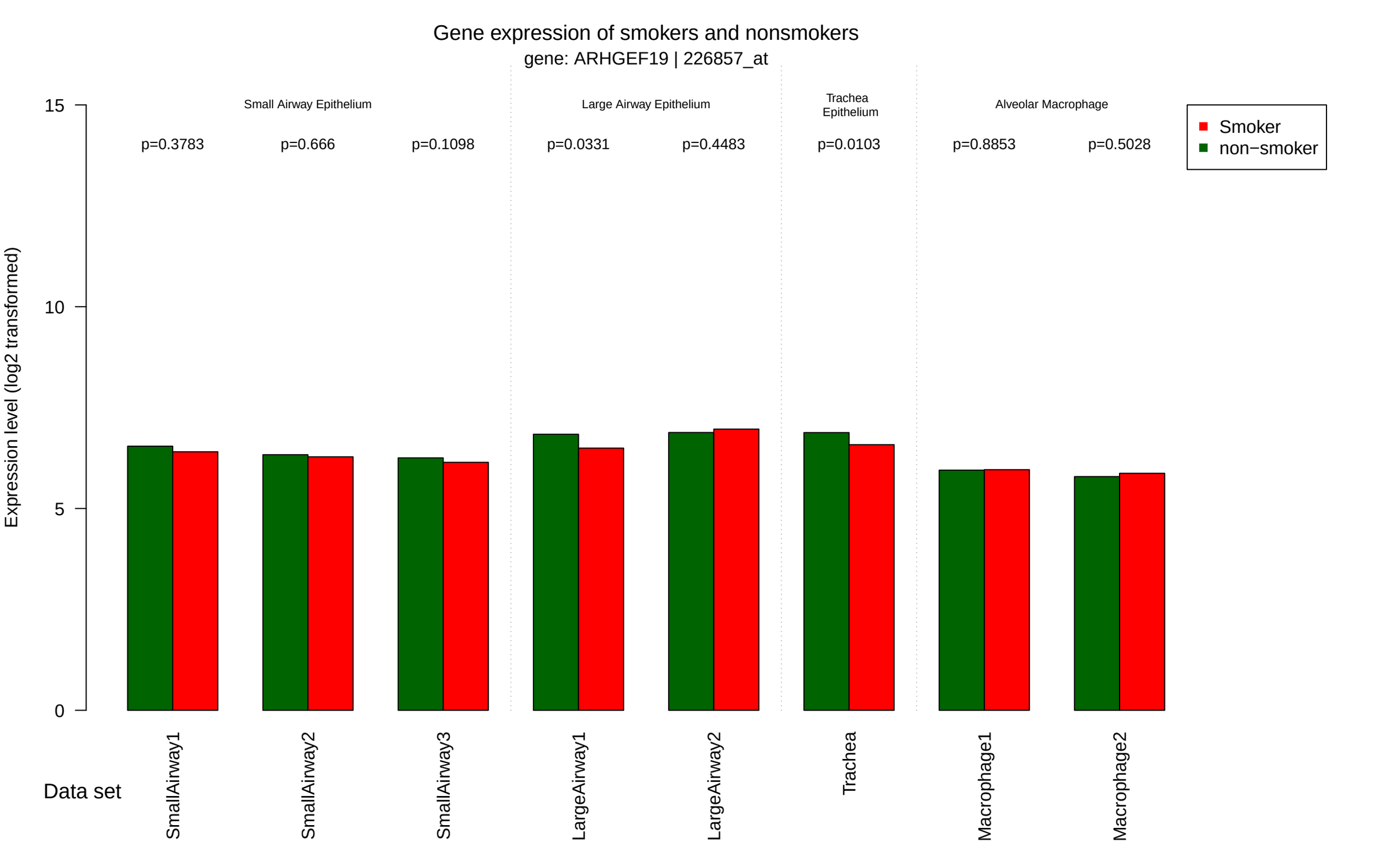
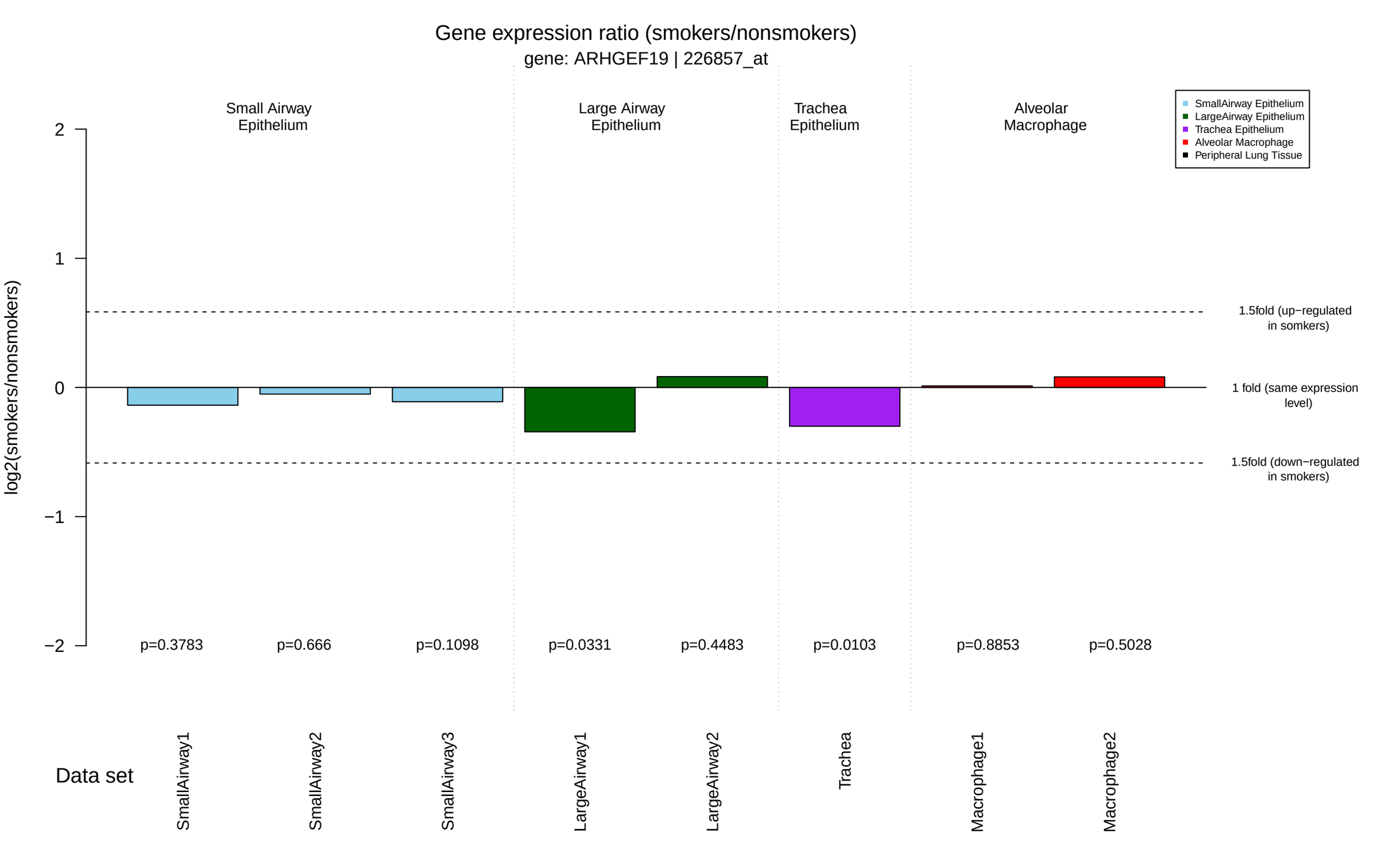
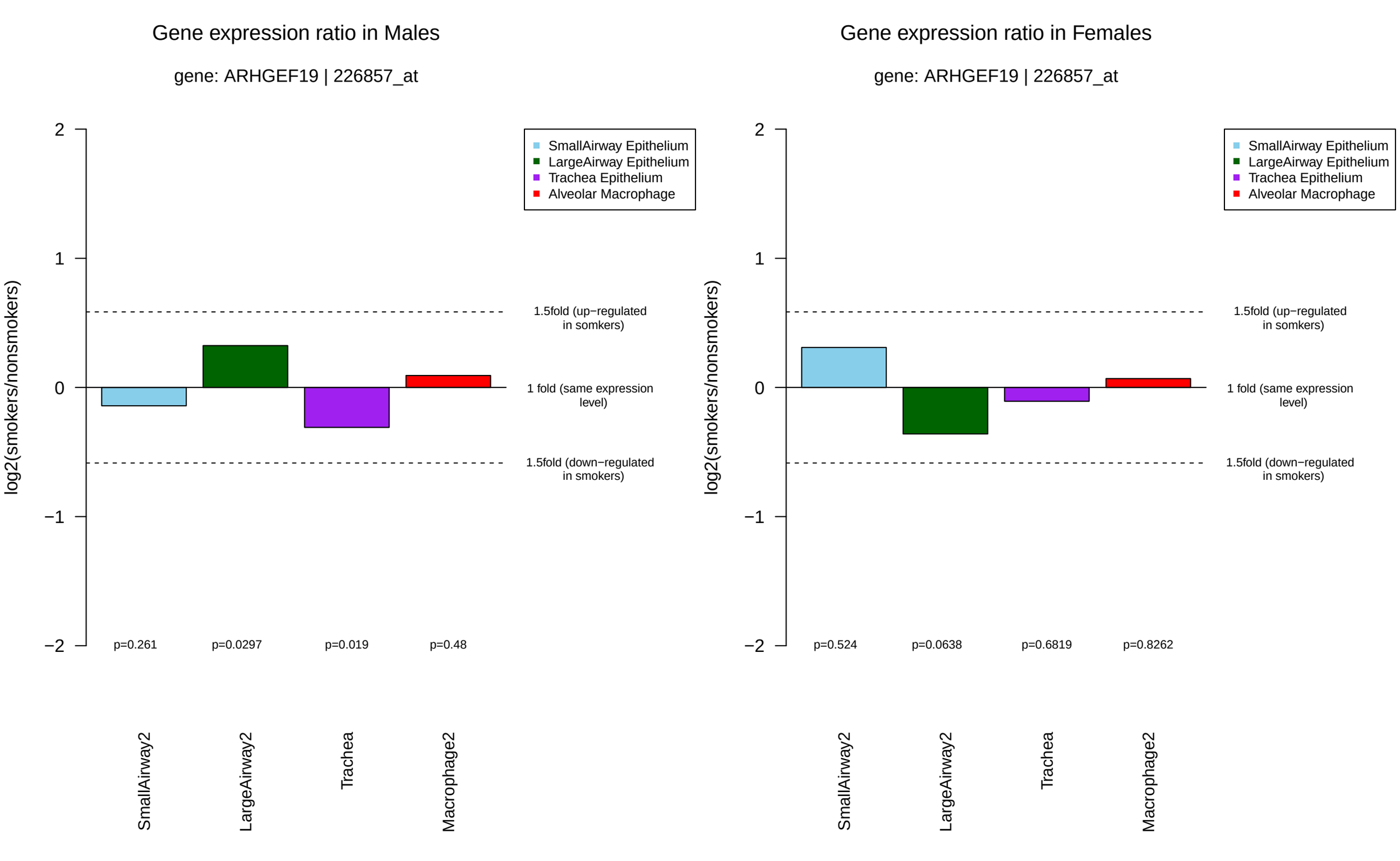
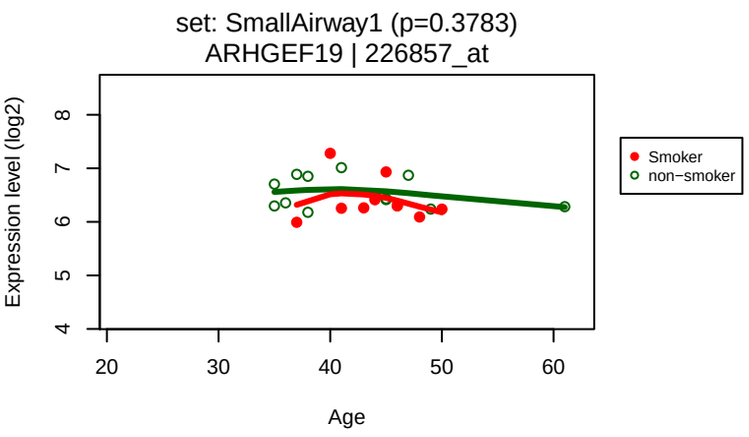
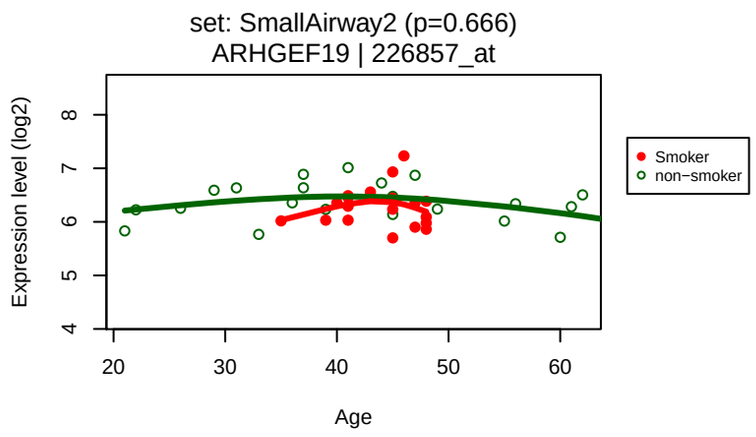
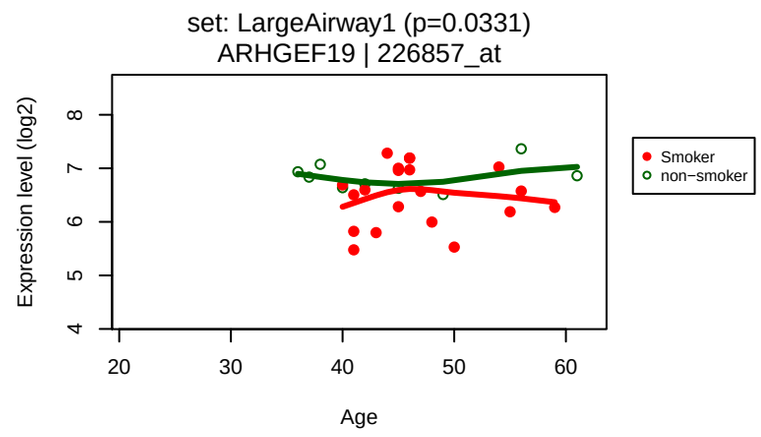

Gene expression of smokers and nonsmokers
gene: ARHGEF19 | 226857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3783
p=0.666
p=0.1098
p=0.0331
p=0.4483
p=0.0103
p=0.8853
p=0.5028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

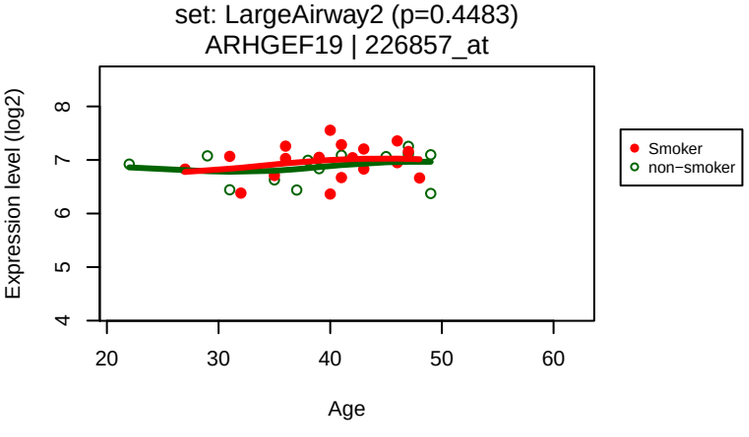

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF19 | 226857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3783
p=0.666
p=0.1098
p=0.0331
p=0.4483
p=0.0103
p=0.8853
p=0.5028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

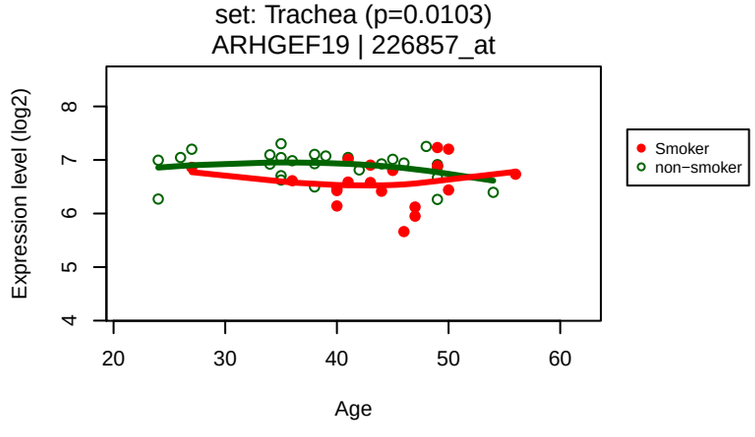

Gene expression ratio in Males
gene: ARHGEF19 | 226857_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.261
p=0.0297
p=0.019
p=0.48
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF19 | 226857_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.524
p=0.0638
p=0.6819
p=0.8262
SmallAirway2
LargeAirway2
Trachea
Macrophage2

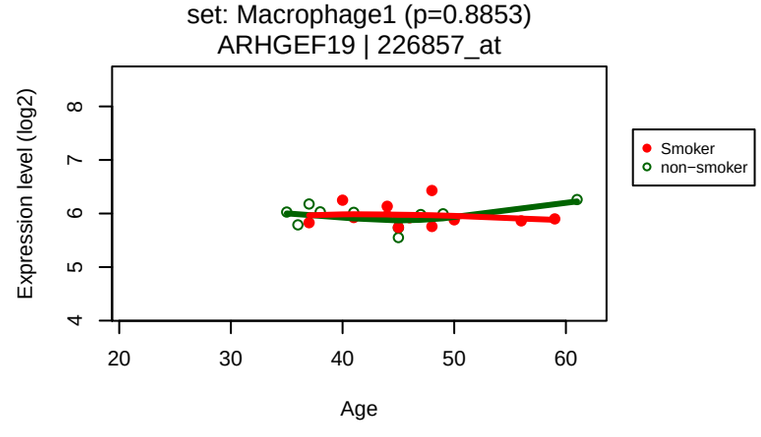

set: SmallAirway1 (p=0.3783)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker

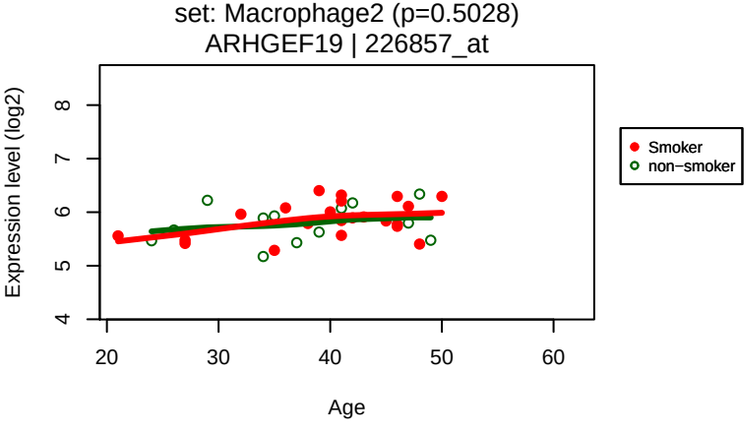

set: SmallAirway2 (p=0.666)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker

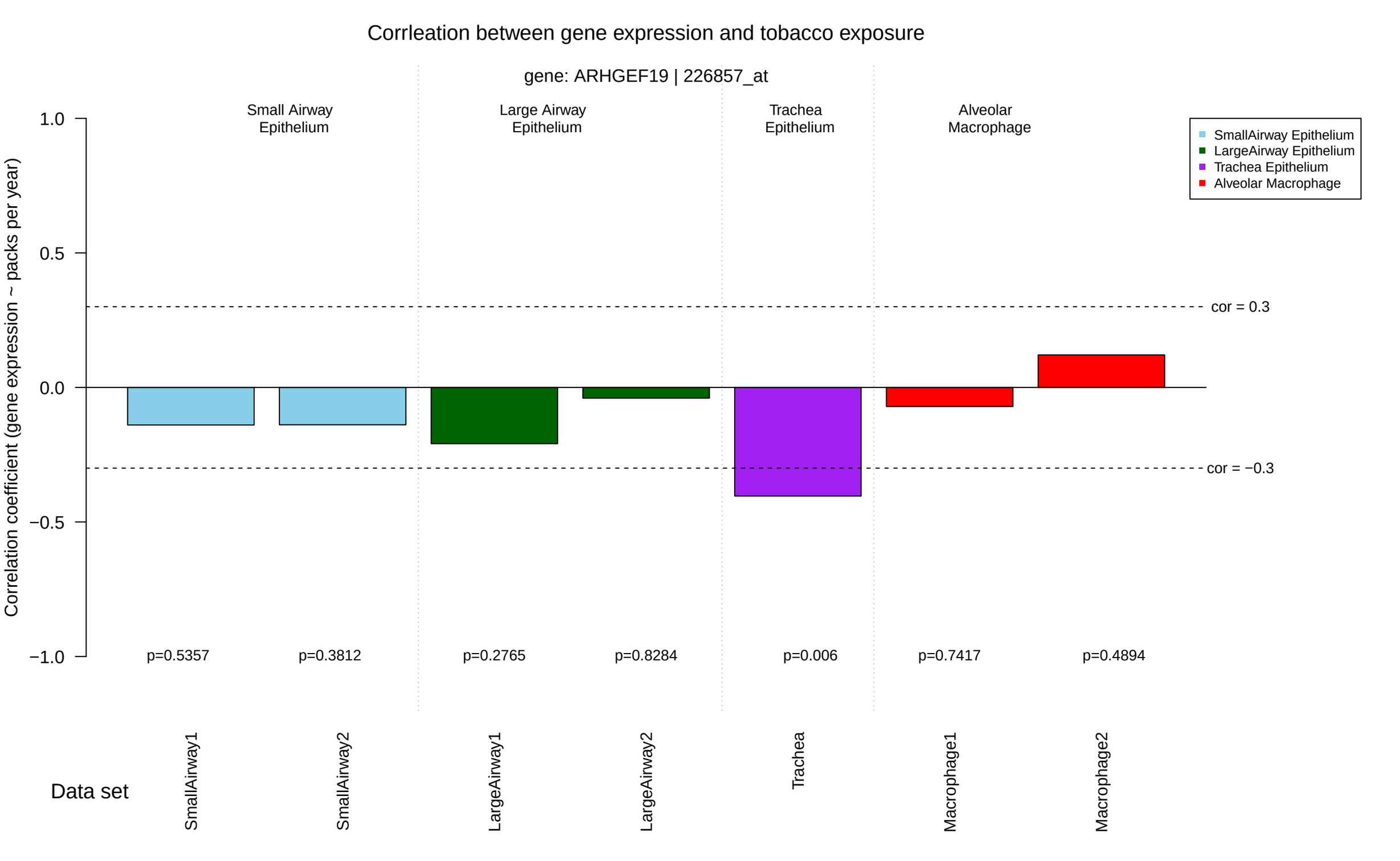

set: LargeAirway1 (p=0.0331)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4483)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0103)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8853)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5028)
ARHGEF19 | 226857_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF19 | 226857_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5357
p=0.3812
p=0.2765
p=0.8284
p=0.006
p=0.7417
p=0.4894
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2